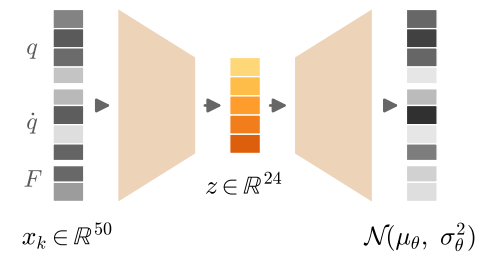


optimize

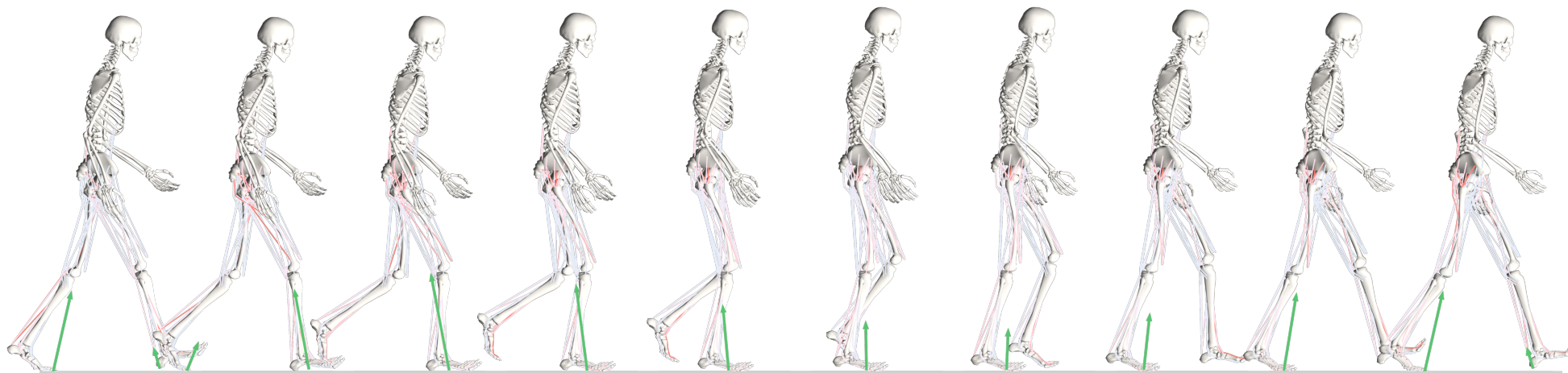
metabolic cost

$$\mathcal{L}_E = \log \frac{1}{md} \int_0^T \dot{E}(x, u) dt$$

state prior



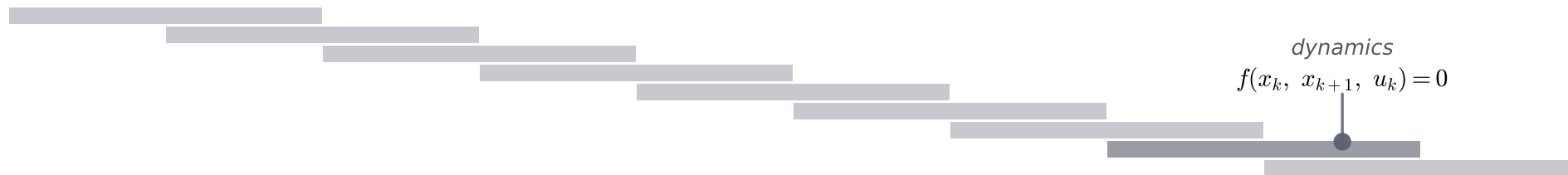
$$\mathcal{L}_{\text{prior}}(x_k) = -\log \mathcal{N}(x_k | \mu_\theta, \sigma_\theta^2)$$



enforce

dynamics

$$f(x_k, x_{k+1}, u_k) = 0$$



$$x_{N+1} = \mathcal{M} x_1$$

half-cycle periodicity